Factor-combination discovery is stochastic threshold group testing: a counterfactual replay of the 2006 reprogramming screen

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October 15, 2026

Abstract

Finding the few transcription factors that drive a cell-fate conversion is expensive: every query is a wet-lab assay, and the 2006 screen for the reprogramming factors consumed a documented 61 under a fixed leave-one-out protocol. We show that this class of screen is an instance of *stochastic threshold group testing*, and replay it on an oracle calibrated to published assay figures. Five results follow. (1) Information-gain design recovers the hidden recipe in a median of 10 experiments against 61, cheaper in 300 of 300 paired replicates; the same inference with random pools needs 44, locating the gain in experiment selection, not inference. (2) An apparent scaling limit, a classical search overtaking the designer at 48 candidates, is a property of the proposals rather than the method: changing only which pools may be scored drops the growth exponent from $n^{1.6}$ to $n^{0.43}$ at 100% recovery, where the classical search falls to 83%. (3) Fixing the recipe cardinality fails *symmetrically and silently*, at 0.999 confidence whether the true recipe is smaller or larger; making size latent removes this for one extra experiment. (4) Audited against two measured perturbation datasets, the assumed gap response of 0.10 should be 0.77; re-run on that structure the designer as calibrated recovers 2.5% of recipes, at mean posterior confidence 0.991 on its failures. (5) That precondition is removable: making the gap response latent recovers 99.3%, at 0.4 experiments above an oracle told its value, and beats a five-experiment calibration pilot on cost and accuracy. The replay validates the procedure, not the oracle, which was calibrated from the screen replayed; only (4) tests it against independent data. Results 3–5 are one finding: every assay parameter such a designer fixes is a silent failure awaiting a system that disagrees, and the remedy is to make it latent rather than calibrate it better.

Author summary

When biologists search for the handful of genes that convert one cell type into another, each guess costs weeks of laboratory work. We asked a counterfactual question about a famous example: had the 2006 screen for the reprogramming factors been planned adaptively, how many experiments would it have taken? On a simulator calibrated to the original paper's numbers, about ten instead of sixty-one. More useful than that number is where the approach breaks. The problem turns out to be a known mathematical one — threshold group testing — which supplies both a theoretical yardstick and a classical algorithm as a competitor. That competitor appeared to win once the candidate gene list passed about forty, which would have confined the adaptive approach to the regime the historical screen occupied; tracing the cause showed instead that the adaptive designer was being offered the wrong experiments to choose among, and supplying the right ones

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restores its advantage at every list length we tested. We also find a failure that matters for anyone deploying these methods: if the number of genes in the recipe is fixed in advance and that number is wrong, the search fails while reporting near-certainty, whether the true recipe is larger or smaller. Finally, we check the model's central assumption against real measurements, find it wrong in the direction that causes exactly this kind of confident failure, and then run the method on the corrected landscape: it recovers the right answer in one run in forty while reporting itself almost certain. The repair turns out to be the same one: stop fixing the number and let the search work it out, which restores the method to full accuracy for less than one extra experiment and beats calibrating it in advance. The general lesson is that an adaptive method is only as trustworthy as the assay model it carries, and that anything such a model treats as known in advance is a confident failure waiting for a biological system that disagrees.

1 Introduction

A cell-fate conversion driven by defined factors is, computationally, a subset-identification problem: of n candidates, an unknown subset R of size k is sufficient, and the task is to find R. What makes it hard is not the size of the space but the price of a query. Each query is a transduction experiment with a biological readout, so budgets are counted in tens.

The canonical instance is the screen of Takahashi and Yamanaka [1], extended to human cells the following year [2]. Twenty-four candidates were selected from the literature; the readout was the number of drug-resistant colonies surviving Fbx15/G418 selection; and the search followed a fixed protocol — each factor alone, the full pool, leave-one-out from the pool, then leave-one-out from the surviving subset — consuming $2n + 13 = 61$ experiments at $n = 24$ regardless of what the data said.

Adaptive experimental design [3–5] should improve on a protocol fixed in advance, and in perturbation biology several groups have shown that it does [6, 7]. What has not been established is the counterfactual for a specific documented discovery: not predictive accuracy over an adaptively collected dataset, but the number of wet-lab experiments needed to reach a known answer under the assay actually used. The 2006 screen is a good vehicle precisely because its experiment count is in the public record.

Our second contribution is a reclassification. The colony assay is not generic combinatorial search: a pool produces colonies efficiently only when it contains all of R, weakly when it contains all but one, and not at all otherwise. In group-testing language [8, 9], a test is negative at or below a lower threshold l, positive at or above an upper threshold u, and stochastic in the gap $g = u - l - 1$. This is threshold group testing [10] in its stochastic form [11], with $u = k$, $l = k - 2$ and $g = 1$. The reclassification is load-bearing rather than decorative: it supplies a theoretical yardstick for the achievable test count and classical algorithms as baselines, and both change the conclusions one would otherwise draw.

Related work. Four literatures meet here, and the closest work in each sits a measurable step away from this one.

Group testing supplies the problem class and its bounds, from the original pooled-screen argument [8] through the modern information-theoretic treatment [9]. The variant that matches this assay is threshold group testing [10] in its stochastic form [11], and designs for it continue to improve [12]. Those designs are parameterised by the threshold u: it is given, and the algorithm is built around it. Here $u = |R|$ is exactly what the experimenter does not know, which is the gap Section 2.5 addresses. Group testing is also being applied to biology right now [13], but in its disjunctive form — a pool is positive if it contains any positive item — and largely non-adaptively,

whereas a reprogramming pool is positive only if it contains *all* of the recipe.

Bayesian experimental design supplies the criterion [3–5]. Applied to perturbation biology, it has been used to choose genetic interventions [6] and combination drug screens [7], and recent work learns an acquisition policy amortised over more than a thousand historical CRISPR screens [14] or selects gene pairs with a predictive model in the loop [15]. These are evaluated on hit-recovery or predictive accuracy at a fixed budget fraction. None asks the counterfactual question of this paper: how many experiments a documented discovery would have taken, measured against the count that discovery actually consumed.

Prediction rather than design is the third line: ranking reprogramming factors from expression and network data [16], scoring transcription-factor combinations from single-cell multiomics [17], and predicting outcomes of unseen multi-gene perturbations [18]. The closest of these even frames its accuracy in experiment-count terms — a cocktail ranked fiftieth implies fifty experiments — but the ranking is static: it does not condition on results as they arrive, and the count is a corollary of the ranking rather than a measured search cost. Such models are nonetheless the natural replacement for the oracle used here, which is the direction the virtual-cell programme points [19].

Validation of adaptive policies is the fourth. Held-out evaluations of selective measurement rules in biological screens have found that the evidence often fails to justify adopting them over a fixed plan [20], a finding that sits naturally beside Section 2.6: efficiency and error detectability pull in opposite directions.

Against that background the contributions of this paper are the counterfactual against a historical experiment count, the regime boundary against a classical algorithm, the treatment of an unknown threshold in a threshold-group-testing assay, and a check of the assay model's central assumption against measured data.

We state plainly what this study is not. The oracle is a simulator calibrated to summary figures, not an assay and not a fit to raw data; Section 2.7 is where measured data enters, and it refutes part of the calibration. No claim about cellular biology follows from any result here.

One inference in particular is unavailable to us, and we flag it because it is the natural reading of a replay. **Recovering the Yamanaka factors is not evidence that the oracle is right.** The oracle was built from the numbers that screen reported, and the recipe is a truth we planted in it; finding that recipe therefore tests the search procedure and nothing else. Treating the recovery as validation of the model would be circular. What the replay measures is the cost of a search strategy under a documented fitness function, against a published experiment count — the same sense in which running a heuristic on a solved instance measures the heuristic, not the instance. Validation of the model happens only once in this paper, in Section 2.7, where its central parameter is compared against measurements it was not fitted to, and fails.

2 Results

2.1 Cost against the historical record

Over 300 replicates (Fig. 1), the historical protocol consumes its fixed 61 experiments; information-gain design needs a median of 10 (95% CI [10, 10]; mean 10.3; worst replicate 18) and recovers the exact recipe in 99.7% of runs; a literature prior reduces this to 6. Paired against the historical protocol, the adaptive designer is cheaper in 300 of 300 replicates.

Two controls matter more than the headline. The ablation needs a median of 44 experiments and recovers the recipe in 88.0% of runs; paired, the difference is 32.7 experiments (permutation $p < 10^{-4}$), with the adaptive designer cheaper in 298 of 300 replicates. The saving comes from informative experiment selection, not from Bayesian updating. The literature prior buys speed

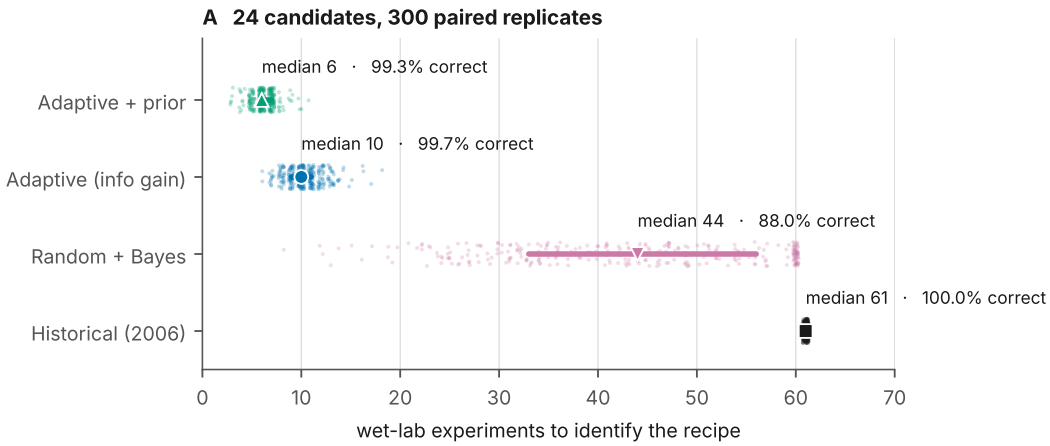


Figure 1: Wet-lab cost of four strategies at 24 candidates, 300 paired replicates. Each point is one replicate; bars span the interquartile range, markers give medians. The ablation — the same exact posterior and stopping rule with pools drawn at random — separates the contribution of experiment selection from that of inference; its pile-up at 60 is the budget ceiling.

and nothing else: -4.1 experiments ($p < 10^{-4}$) with no detectable accuracy difference (McNemar $p = 1.0$).

The measured cost should be read against theory rather than against the 2006 protocol alone. For stochastic threshold group testing, test counts of order $g^2 d \log n$ are achievable [11]; at $g = 1$, $d = 4$, $n = 24$ that is ≈ 13, consistent with the observed median of 10. The 84% saving is therefore not a surprising empirical finding but the expected consequence of replacing a $\Theta(n)$ protocol with one operating at the model's known order. That is itself the point: the gap between what the 2006 screen spent and what was achievable was derivable from theory contemporaneous with it.

2.2 An apparent regime boundary

Yamanaka's 24 candidates were themselves the product of a literature filter, so a method that works only after such filtering is of limited use. Figure 2 varies the list from 24 to 60 factors, i.e. hypothesis spaces from 10,626 to 487,635.

The Bayesian designer's wet-lab cost grows as $n^{1.8}$, not logarithmically, and its advantage over the historical protocol decays from 84% to 57%. The classical half-removal search shows the opposite pattern: twice as expensive at $n = 24$ (20 versus 10 experiments), it grows as $n^{0.4}$ and the two curves cross at $n = 48$ (26 versus 27). Because the classical search carries no model of the assay, this localises the deficiency: not in the inference and not in the problem, but in proposal generation. Random pools of size 2–12 contain all four essential factors with probability falling as n^{-4}, so as the list grows the designer evaluates information gain ever more accurately over proposals ever less likely to be informative. The remedy is a better proposal distribution, and the classical algorithm is in effect one — a diagnosis we test directly in Section 2.3, where it proves correct and the boundary disappears.

Neither method dominates. The classical search pays for its simplicity in reliability: its success rate falls from 100% to 83% as n grows to 60, for a physical reason rather than an algorithmic one. The pool-size penalty drags a full 60-factor pool down to about 6 expected colonies, so the positive control the algorithm starts from approaches the noise floor. The Bayesian designer holds 100% at every size because it reasons from the whole count distribution rather than from one threshold.

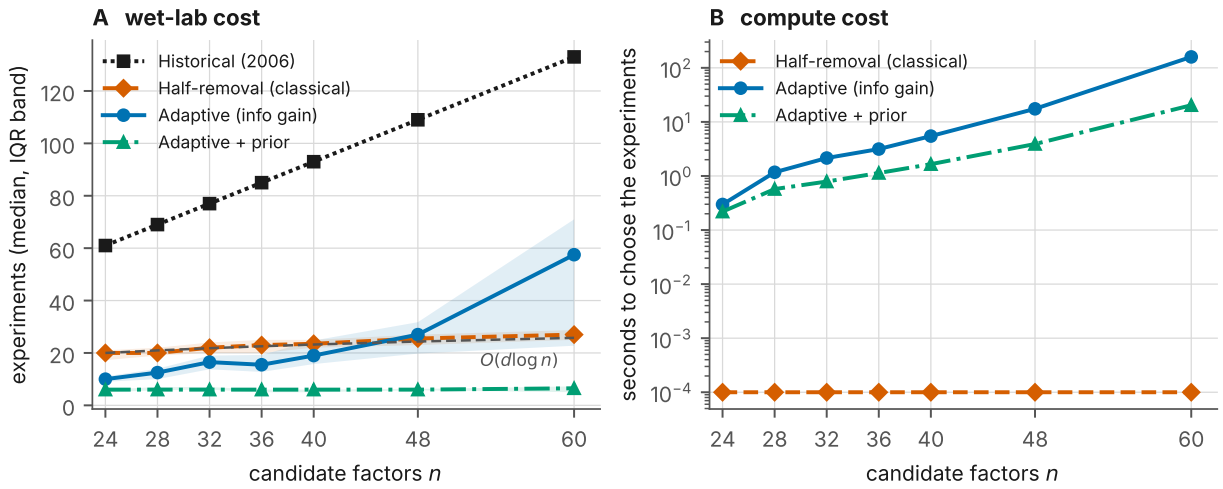


Figure 2: Growth with candidate-list size. (A) Wet-lab cost, medians with interquartile bands, against the $O(d \log n)$ reference curve scaled at the smallest list. (B) Compute spent choosing experiments, log scale. The classical half-removal search recovers logarithmic growth at negligible compute; the Bayesian designer does not, and the two cross at 48 candidates.

Compute separates them further: 159 s per run at $n = 60$ against under a millisecond.

The $k + 1$ overlap-class collapse (Section 4.2) makes the cost of scoring a proposal independent of posterior concentration, but it remains linear in the enumerated space, and the number of rounds grows too; empirically design cost grows as $H^{1.5}$ in the hypothesis count H. Exact enumeration is the binding constraint beyond about 60 candidates. One curve in Figure 2 must be read with care: the literature-prior variant appears flat in n only because prior quality is held constant, up-weighting eight factors regardless of n. A longer candidate list implies a less informative prior, so that curve is an upper bound conditional on prior quality, not a prediction.

2.3 The boundary is a property of the proposals, not the method

Section 2.2 attributed the adaptive designer's poor scaling to its proposal distribution rather than to its inference. That attribution was an interpretation of a null result and deserved a direct test, since a method's authors are poorly placed to decide that its deficiency is the benign kind.

We therefore constructed a second proposal generator and changed nothing else. It adds the three families the original lacked. *Removal pools*: the factors carrying appreciable posterior mass, less a random half, quarter or eighth of them — precisely the queries by which half-removal halves its live set, and absent from the original proposals at every list length. *Complements of ranked prefixes*: a prefix of the marginal ranking asks whether the recipe lies in the top k, and its complement asks the same question from the side that is more informative when the answer is no. *Scale-aware random pools*: sizes drawn log-uniformly between the recipe size and the full candidate set, rather than from a fixed window of 2–12. Both generators return the same number of pools per round, so no difference can be bought with a wider search.

The diagnosis holds (Fig. 3). The growth exponent falls from $n^{1.61}$ to $n^{0.43}$, against $n^{0.36}$ for the classical search, and the median cost at 60 candidates falls from 48 experiments to 15 — below the classical algorithm's 27 rather than above it. The saving is zero at 24 candidates (0.6 experiments, $p = 0.29$), where the original window already covered a usable fraction of the space, and grows monotonically with list length: 5.5 at 32, 8.6 at 40, 19.7 at 48 and 34.2 at 60 experiments per run,

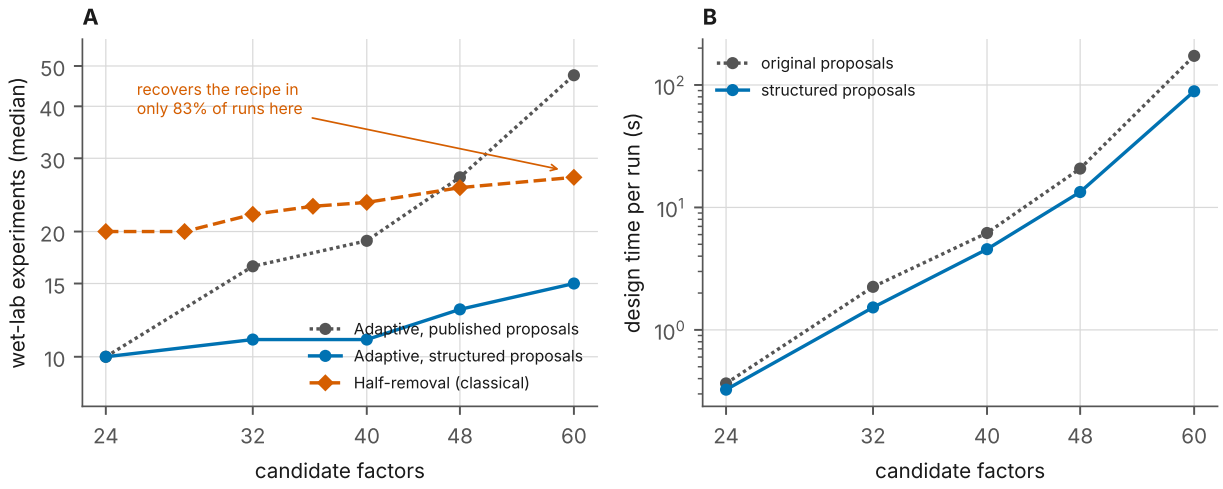


Figure 3: Changing only which pools may be scored. (A) Median wet-lab cost against candidate-list size for the published proposal generator, the structured one, and the classical half-removal search of Section 2.2. The scoring rule, posterior, oracle, recipes and seeds are identical across the two adaptive curves, as is the number of pools scored per round. (B) The design-time price of the change, which is negative: the structured proposals need fewer rounds than their larger pools cost.

all paired on identical recipes and seeds, all $p < 10^{-3}$. Recovery remains 100% at every size, where the classical search declines to 83% at 60 for the physical reason given above.

Two qualifications. The comparison is confined to the hypothesis spaces exact enumeration can hold, so it does not show that the Bayesian designer scales indefinitely; the compute wall of Section 2.2 is untouched, and remains the binding constraint past about 60 candidates. And the structured generator was designed after seeing the failure it repairs, which makes it an explanation of that failure rather than an independent prediction. What the experiment establishes is narrower and sufficient: the regime boundary we reported was contingent on an implementation choice, not a property of information-gain design, and should not be cited as the latter.

2.4 Model error: absent signal versus wrong signal

The graded three-of-four signal is the gap response of the threshold model, and it is natural to suppose the search depends on it. Separating the signal's presence from the searcher's belief about it shows otherwise (Fig. 4A). With the gap response removed entirely ($\rho = 0$) and the likelihood informed of this, the recipe is still recovered in 100% of runs at a median of 22 experiments: the search degenerates to pure group testing, pays a factor of 2.2 for the lost gradient, and remains 64% cheaper than the historical protocol. The same oracle with a searcher that keeps believing $\rho = 0.10$ succeeds in 52% of runs. Absence of graded synergy is a calibration error, not a structural barrier.

What binds is the magnitude of the error, asymmetrically. Under-estimating the gap response fivefold still recovers the recipe in 70% of runs, while over-estimating it fourfold drops recovery to 42% — and reports confidence 0.989 while doing so. Conservative calibration is the safer error direction, which matters because the gap response is exactly the quantity a pilot experiment estimates poorly.

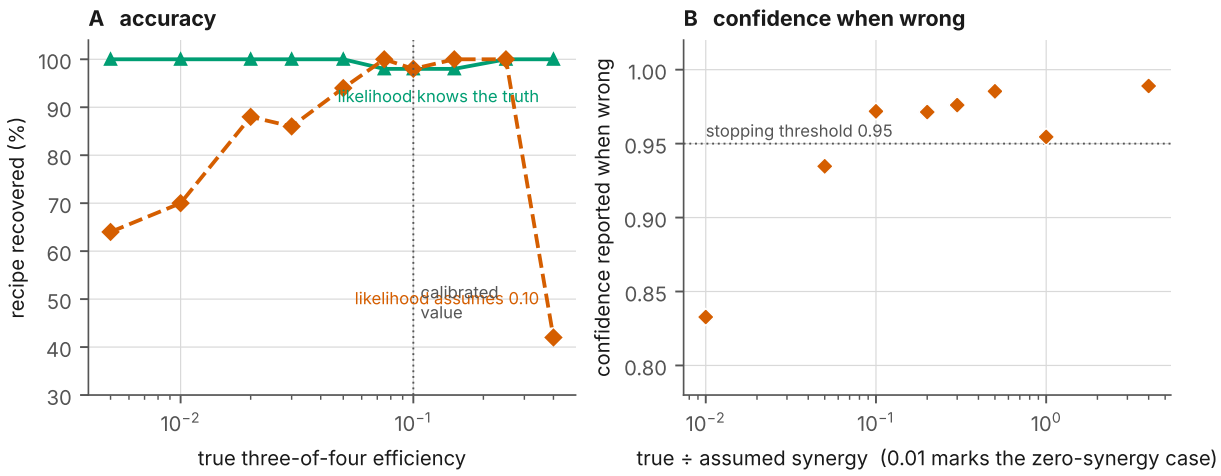


Figure 4: Gap response and calibration error. (A) Recovery rate as the true three-of-four efficiency ρ varies, with the likelihood either informed of it or holding the calibrated 0.10. (B) Confidence reported in runs that returned a wrong recipe, against the ratio of true to assumed ρ (the leftmost point is the zero-gap case).

2.5 Assumed cardinality: a symmetric, silent failure

Every result above assumes the searcher knows $k = 4$. Relaxing that assumption exposes the method's one genuinely structural failure (Fig. 5). A designer fixed at four factors recovers a three-factor recipe in 0% of runs and a five-factor recipe in 0% of runs, and in both cases reports a mean posterior confidence at or above 0.99 on the wrong answer. The failure is symmetric: searching for more factors than exist is as fatal as searching for fewer, because the fixed designer appends a spurious factor to a correct core and then concludes. This is not a failure of the data but of a hypothesis space that cannot represent the truth, and the posterior is confident precisely because every hypothesis it can represent is wrong by the same margin.

Treating cardinality as latent removes it. Recovery is 100% at true sizes three, four and five; the correct size carries posterior mass 0.995 or above in all three cases; and the premium on the four-factor case the fixed designer was built for is +0.9 experiments (paired $p = 0.0036$). The latent treatment also returns a diagnostic the fixed designer cannot produce even in principle: a posterior over recipe size, which is what would tell an experimenter that their assumed cardinality was wrong.

Two comparators place this result. The classical answer to unknown $|R|$ is to estimate the count in a first stage and then commit to it [21]. That baseline inherits the failure whenever the first stage errs, and it errs often: with a six-experiment first stage it commits to the wrong size in 53% of four-factor runs and 87% of five-factor runs, recovering the recipe in 47% and 13% of them; doubling the first stage to twelve experiments reduces the wrong commitments to 7% and 40%, for 93% and 60% recovery, at a cost of +1.7 experiments on the four-factor case ($p = 0.005$). Carrying the size posterior throughout instead of committing to it is thus not redundant machinery: it is what converts a 60% method into a 100% one at no measurable premium (+0.5 experiments, $p = 0.53$). The model-free half-removal search sidesteps the question entirely, since it never represents a size, and recovers all three sizes without modification — for +9.7 experiments ($p < 10^{-4}$), roughly double. Cardinality is therefore a liability of the parametric approach specifically, not of the problem.

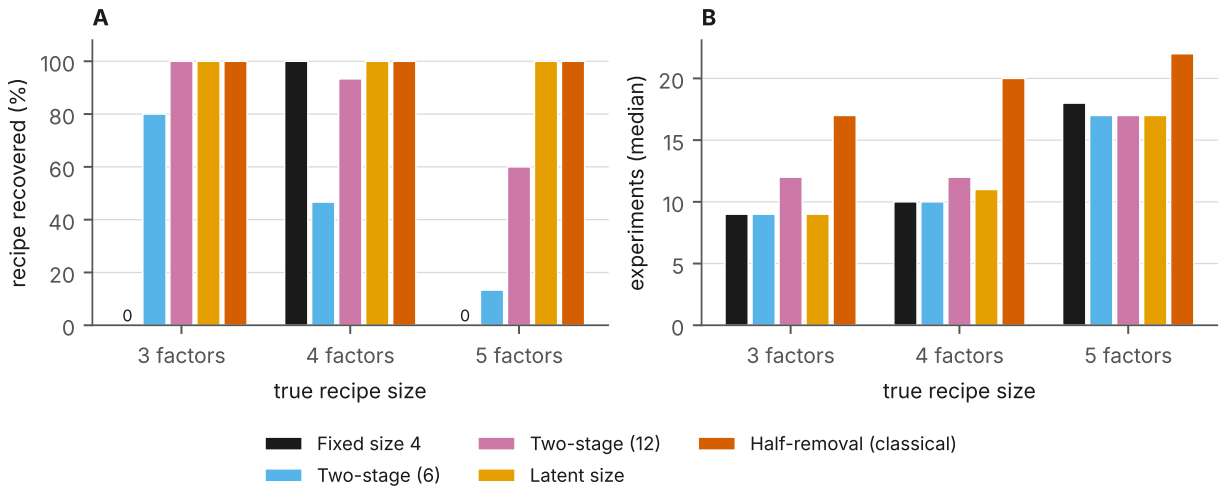


Figure 5: Unknown recipe size, several treatments, three truths. (A) Exact-recipe recovery. (B) Cost. Fixing the size at four fails completely when the truth is three or five; estimating the size in a first stage and committing to it inherits the failure whenever the estimate is wrong; carrying size as a latent variable, or dispensing with a size representation altogether, does not.

2.6 The stopping rule, and a trade-off between efficiency and detectability

A Bayesian stopping rule is usable only if its confidence means what it says, since in a real campaign there is nothing to check it against. Sweeping the threshold from 0.50 to 0.9999 (Fig. 6), the stated confidence never exceeds the observed recovery rate by more than its Wilson interval permits (largest discrepancy +0.025), and tightening by four orders of magnitude costs 0.8 experiments on average while never moving the median. The practical recommendation is to run at 0.999 or tighter.

That flatness is the mechanism behind a pattern visible throughout. Confidence does not accumulate gradually; it jumps at the decisive experiment, so there is no intermediate regime in which the searcher can report that it is not sure yet. When the model is wrong, high confidence therefore arrives just as promptly on the wrong answer: the adaptive designer's mean confidence when wrong is 0.955 against 0.423 for the random-pool ablation; an over-estimated gap response fails at 0.989; a cardinality mismatch fails at 0.999. The designer is well calibrated *inside its model*, and that does not protect it when the model is wrong. Efficient experiment selection improves cost and degrades the detectability of model error at the same time, because a procedure that gathers consistent evidence quickly gathers it quickly for a wrong hypothesis too.

2.7 The gap assumption against measured data

Every result above is conditional on gap structure of the assumed shape existing. Single-cell perturbation screens now cover enough combinations to check it directly [24, 25], and two of them carry the pairs and singles the measurement needs. For a perturbation p with pseudobulk profile x_p and reference x_0, write $\delta_p = x_p - x_0$; for each combination whose one-short constituents were also profiled, we measure

$$\text{one-short} = \frac{\max_c \|\delta_c\|}{\|\delta_{\text{combination}}\|}, \tag{1}$$

the empirical analogue of ρ, whose calibrated value is 0.10.

In Norman et al. [22] — K562, CRISPRa, 131 pairs with both singles measured — the median

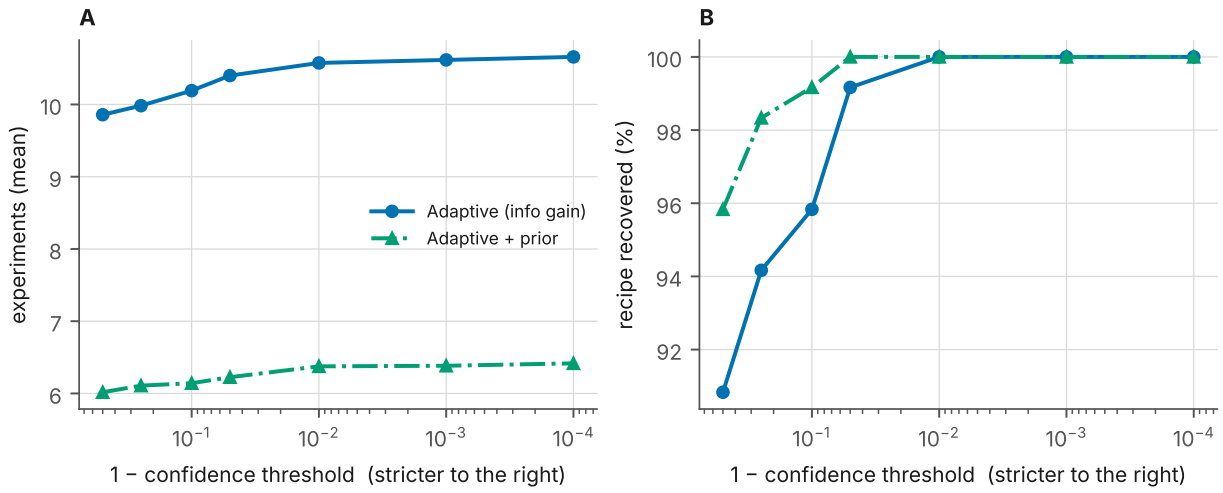


Figure 6: The price of certainty. (A) Mean experiments and (B) recovery rate as the stopping threshold tightens from 0.50 to 0.9999. Cost is nearly flat while accuracy improves, because posterior confidence jumps at the decisive experiment rather than climbing gradually.

is **0.77** (IQR 0.62–1.03), and **no pair of the 131** falls below 0.20. In the combinatorial subset of Joung et al. [23] — human embryonic stem cells, transcription-factor overexpression, 40 TF–TF pairs — the median is **0.77** (IQR 0.52–0.91) under a reference fitted by least squares, 0.91 and 0.82 under two alternative references, with no pair below 0.20 under any of them (Fig. 7B). The two measured three-factor combinations give 0.80 and 1.01.

The result is not an artefact of measuring effect size by a Euclidean norm. Repeating the Norman analysis with threshold-like functionals — the number of genes moving more than 0.25 or 0.5 in log space — lowers the median to 0.48 and 0.50 and raises the fraction below 0.20 to 25% and 22%, but no functional produces a median near 0.10.

Two further observations. First, departures from additivity are common but modest in magnitude: in Norman the median ratio of the pair's effect to the sum of its singles is 0.91, with 47% of pairs sub-additive and 27% super-additive, and only 28% of pairs have an additivity residual exceeding its own sampling noise, estimated by regressing squared residuals on $1/n_{AB} + 1/n_A + 1/n_B$. Second, the two systems differ in *where* the interaction lives: the median R^2 of a pair on its two singles is 0.86 in K562 but 0.64 in hESCs, so combinations in the stem-cell system leave the plane spanned by their constituents more, while their magnitudes remain near-additive.

The implication is specific. Our calibrated $\rho = 0.10$ is five- to eightfold sharper than measurement supports, and Section 2.4 shows that over-estimating the gap response is precisely the direction in which the designer fails silently. A designer calibrated as ours is would therefore be expected to fail confidently on these systems, while the same designer informed of the true ρ succeeds in 98–100% of runs at every value tested, including $\rho = 0$. The gap response must be estimated, not assumed.

The constructive reading is that the oracle should be replaced rather than recalibrated: a predictive model fitted to measured perturbation outcomes [18], or the broader virtual-cell programme [19], would supply a gap response estimated from data rather than assumed, and the designer would inherit whatever accuracy that model has. The experiments above then become the specification for how accurate it must be: within a factor of two on the gap response, and honest about recipe size.

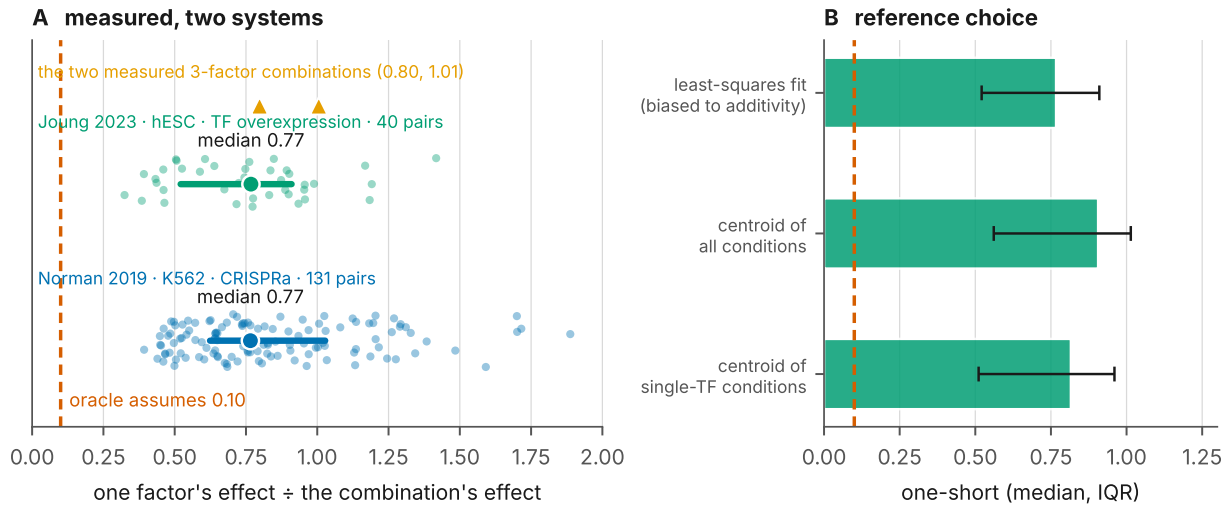


Figure 7: What one factor of a measured combination reproduces. (A) The ratio $\|\delta_{\text{one short}}\|/\|\delta_{\text{combination}}\|$ for 131 gene pairs in K562 [22] and 40 transcription-factor pairs in human embryonic stem cells [23], with the two measured three-factor combinations marked. The oracle assumes 0.10. (B) The hESC subset has no unperturbed control, so three references are reported; the conclusion does not depend on the choice.

One boundary deserves emphasis. Both datasets read out the transcriptome, while the 2006 assay reads out a rare discrete phenotype — a colony forms or does not — which is a thresholded functional of cell state and may well be far sharper than transcriptome displacement. The audit therefore refutes the assumption for expression-level objectives and constrains, rather than settles, the phenotype case. Settling it requires a combinatorial screen with a phenotypic readout, which to our knowledge does not exist at the scale needed.

2.8 The designer run on the measured gap structure

Section 2.7 established that $\rho = 0.10$ is far sharper than measurement supports, and inferred from Section 2.4 that a designer calibrated this way *would be expected* to fail confidently. An inference of that weight should be measured rather than inferred. We therefore re-ran the search with ρ drawn per replicate from the 131 measured Norman ratios and held fixed across arms, so that every method meets the same landscape (Fig. 8).

The expectation is confirmed, and more sharply than anticipated. Under the gap fall-off the designer as published recovers the recipe in **2.5%** of 200 runs (95% Wilson interval 1–6%) against 99.0% for the same designer informed of ρ (exact McNemar $p < 10^{-4}$). The failures are not budget exhaustions: no run reached the cap, the median cost was 9 experiments against 10 for the correctly calibrated reference, and the mean posterior confidence on runs that ended *wrong* was **0.991**. The practitioner receives a four-factor recipe, a tight posterior and no indication of error.

Three consequences follow. First, the fault is in the calibration rather than the inference: the matched designer costs 12 experiments against 10 at the assumed ρ, so the headline saving survives the realistic landscape *conditional on knowing* ρ. Estimating the gap response is therefore not a refinement but a precondition, and the natural construction — carrying ρ as a latent parameter updated alongside the recipe — is the obvious next step rather than an optional one.

Second, the classical algorithm is not a fallback. Half-removal recovers the recipe in 0.0% of

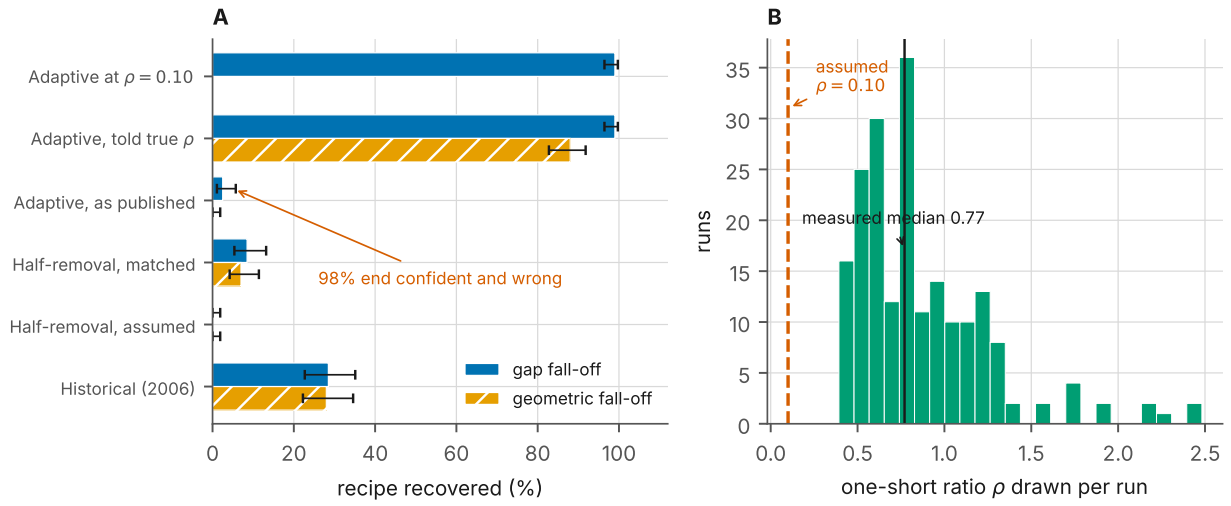


Figure 8: Recovery when the gap response is drawn from measurement rather than assumed. (A) Recipe recovered over 200 paired runs per arm, with Wilson intervals, under both fall-off shapes. The designer carrying this study's published calibration recovers the recipe in 2.5% of runs and stops on its own confidence rule in almost all of the failures. Told the true ρ, the same designer returns to 99%. (B) The distribution of ρ drawn per run, against the value the oracle assumes.

runs with its assumed threshold and 8.5% with the threshold rebuilt at the true ρ: when a pool one factor short yields 80% of a complete one, no positivity threshold separates them, and being told the truth does not help because the information is not in the readout. The 2006 protocol reaches 28.5%, worse than the matched designer by a wide margin but without a confidence claim to be wrong about — it fails audibly.

Third, the conclusion is not an artefact of the sharp fall-off. Under a geometric shape, where efficiency decays as ρ per missing factor rather than vanishing beyond one, the published designer recovers nothing at all (0/200) while the matched designer reaches 88.0% at a median of 16 experiments and a mean of 35. Pair measurements constrain the one-short step and say nothing about what lies beyond it, so both shapes are reported.

2.9 Discovery against measured data, with no simulator

Experiment 2.8 grounds one parameter of the oracle in measurement, but still queries a simulator. The strongest available test removes it. We built an oracle over Norman et al. [22] that answers only for conditions the authors actually profiled — 105 singles, 131 pairs and one control — returning the magnitude of the measured pseudobulk shift against control, perturbed by the dataset's own disagreement across two random halves of the gene panel (median relative discrepancy 0.049). A condition that was not profiled raises an error rather than being interpolated, since an interpolated answer would reinstate the simulation the construction exists to remove.

That restriction changes the task. With no pools beyond pairs there is nothing to design, so this is sequential discovery within a measured library rather than threshold group testing: find the strongest combination while profiling as few conditions as possible. It tests the half of the claim that survives — that choosing the next experiment from a model beats choosing it by protocol.

The strongest measured combination is `CEBPA+JUN`. A surrogate fitting pair response on its constituents' measured singles, queried by upper-confidence score, reaches it in a median of 68 conditions in 40 of 40 replicates, against 108 for a singles-first protocol and 84 for random order,

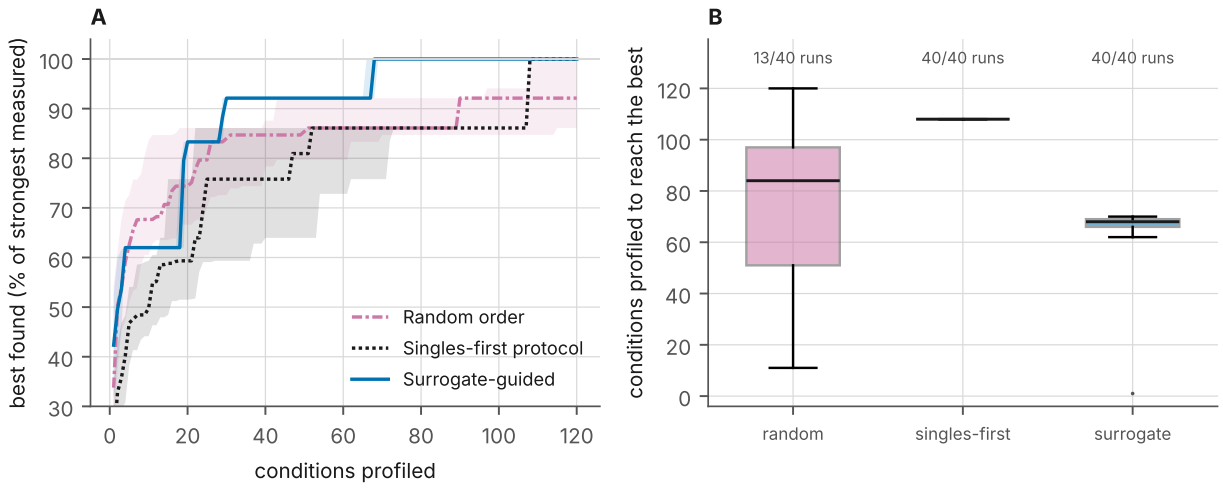


Figure 9: Sequential discovery where every query returns a measurement. (A) Best condition found so far as a fraction of the strongest measured combination, median over 40 replicates with interquartile band. (B) Conditions profiled before reaching that combination; the annotation gives how many replicates reached it at all, without which the box is not comparable.

which reached it at all in only 13 of 40. At a budget of 20 conditions the surrogate attains 83% of the target against 59% for the protocol (paired permutation, $p < 10^{-4}$; Fig. 9).

The singles-first row should be read as biology before method. A measured one-short ratio of 0.77 is precisely what makes "profile the singles, then the pairs built from the best of them" nearly sufficient on this landscape. The simulated assay, in which a single factor yields nothing at all, is the *easier* of the two problems rather than the harder one — a point that the simulation-only version of this study could not have reached.

2.10 Inferring the gap response rather than assuming it

Section 2.8 leaves ρ as something that must be obtained rather than assumed. A laboratory would obtain it by calibrating first: run a recipe already known to work and each of its one-short pools, take the ratio, then fix ρ and search. The alternative is to carry ρ in the posterior and let the search inform it. Which is cheaper is not obvious — the pilot spends whole experiments up front and ends in a point estimate with its own Poisson error, while joint inference spends part of every experiment's evidence on a parameter nobody asked about.

The joint posterior is the fixed-size designer with one more index. The colony distribution depends on a hypothesis and an assay model only through the pair $(|S \cap h|, m)$, so the five overlap classes become $5 \times |\mathcal{M}|$ classes and the cost of scoring a proposal again does not depend on how concentrated the posterior is. We discretise ρ on nine points spanning the assumed 0.10, the measured range and the sub-additive region above 1, and stop on the *recipe* marginal with ρ integrated out, since a designer that waited to pin down a nuisance parameter would be answering the wrong question. All arms use the structured proposals of Section 2.3, so the generator is not a confound.

Under the gap fall-off the result is clean (Fig. 10). Latent ρ recovers the recipe in **99.3%** of runs at a median of 12 experiments — indistinguishable from the oracle arm told the true value (99.3%, median 12), at a cost of 0.4 experiments on the paired mean. Against the pilot it is both cheaper and more accurate: 4.7 fewer experiments ($p < 10^{-4}$) at 99.3% against 95.3%. Making the shape latent as well costs nothing measurable and recovers 100% of recipes. The practical recommendation is

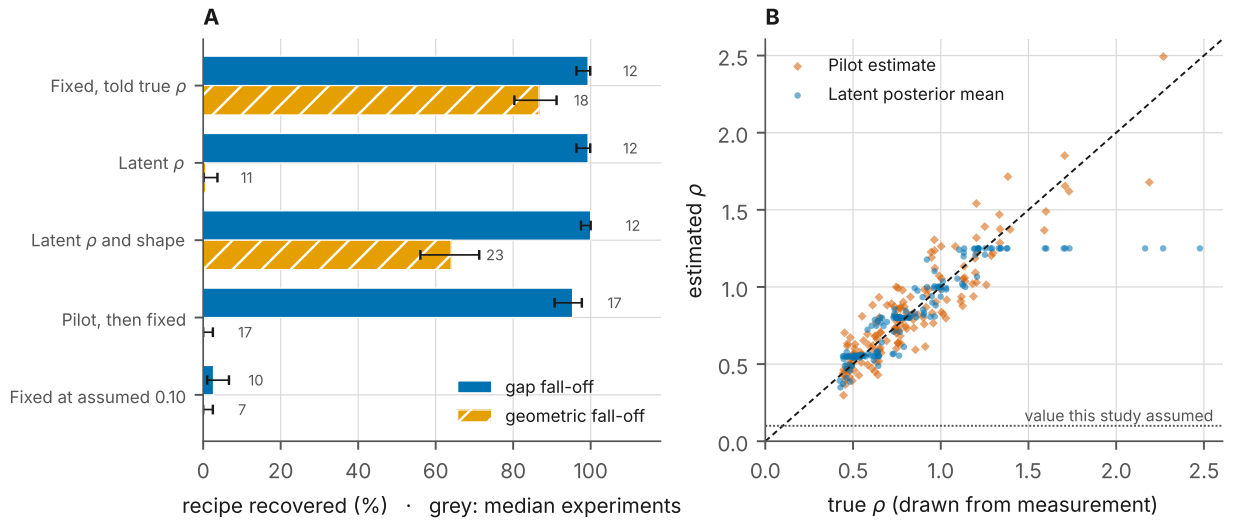


Figure 10: Two ways to stop assuming ρ. (A) Recipe recovered over 150 paired runs per arm with Wilson intervals, under both true fall-off shapes; the grey number beside each bar is that arm's median experiment count, since an arm is only cheap if it is also right. The pilot's five calibration experiments are counted inside its total. (B) Estimated against true ρ for the pilot and for the latent posterior's mean, under the gap shape, with the identity line. The latent estimates flatten at 1.25 because that is the top of the grid; the pilot has no ceiling but is noisier throughout, and both are far from the 0.10 this study assumed.

therefore the stronger of the two available: ρ need not be calibrated in advance at all, and a designer that infers it is no more expensive than one handed the answer.

The geometric fall-off is where the result stops, and the way it stops is informative. Latent ρ with the shape still fixed recovers **0.7%** — no better than the uncalibrated designer. The pilot recovers **0%**, and necessarily so: a pilot measures the one-short step and is structurally blind to what lies beyond it, so no number it returns can repair a wrong shape. Only the arm that treats the shape as unknown recovers anything, at 64.0% and a median of 23 experiments, against 86.7% for an oracle told both shape and ρ. That arm identifies the shape essentially perfectly — posterior mass on the true shape above 0.9 in 100% of runs — so what limits it is not identifying the model but searching the flat landscape the model describes, which Section 2.8 already showed to be hard even with full knowledge. It also fails silently in 18% of runs, which is worse than any arm under the gap shape.

The inferred ρ is accurate where the shape is right and biased where it is not: median absolute error 0.06 against a grid spacing of 0.20–0.25 under the gap shape, with the 90% credible interval over grid cells covering the true value in 86% of runs — close to nominal, and a little under it, which is the expected price of a discrete grid. Under a misspecified shape the same interval covers in 36% of runs, so the ρ posterior reports the miss rather than concealing it.

Two things follow. The parameter this study got wrong is now inferable at no cost, which removes the precondition of Section 2.8. But the broader lesson of that section survives intact: the fix generalises only as far as the model class does, and a designer that fixes the fall-off *shape* inherits exactly the failure that fixing ρ produced. Measured pair data cannot distinguish the shapes, so this is not a gap we can close with the data in hand; it is the next measurement someone has to make.

3 Discussion

Five conclusions survive the controls, and they are narrower than the headline saving.

The saving is real but unsurprising once the problem is named. Recognising the colony assay as stochastic threshold group testing with $g = 1$ supplies the achievable order, $O(g^2 d \log n)$, and the observation that the 2006 protocol operated at $\Theta(n)$. The contribution is not the adaptive designer but the quantification: how much of that theoretical gap is realisable under the assay's noise, with the historical count as the reference.

The Bayesian apparatus does earn its place, but the reason is not the one we first reported. Measured against a classical conjunctive search it appeared to lose beyond 48 candidates; giving it the classical algorithm's own queries to score removes the crossing entirely and leaves it at a third of the cost and a higher recovery rate at every list length tested (Section 2.3). The episode is worth stating plainly, because the first version of this analysis would have been published as a regime claim and the regime was an artefact. What survives is a compute boundary rather than a statistical one: the design step still costs 89 s per run at 60 candidates against under a millisecond for the classical search, and exact enumeration remains the binding constraint.

For deployment, the cardinality result is the most actionable: a parametric designer that fixes recipe size fails silently and symmetrically, the standard two-stage remedy inherits the failure, and latent-cardinality inference removes it for about one extra experiment while supplying a size posterior that makes the error visible. Given the general pattern that efficient design suppresses the signature of model error, cheap diagnostics of this kind are not optional extras.

Finally, the real-data audit turns the study's main caveat into a result, and then into a verdict on the study's own artefact. The assumption that a one-short pool yields a tenth of the full response is not supported in either measured system; the measured value is close to 0.8. Running the designer on that structure rather than reasoning about it (Section 2.8) gives 2.5% recovery at a mean posterior confidence of 0.991 on the failures. The operational rule follows without hedging: measure the gap response in a pilot, calibrate conservatively, and treat a designer that has not done so as untrustworthy regardless of what its posterior reports — including the one distributed with this paper, which should be run with ρ estimated and not with the value we calibrated.

Section 2.10 then removes that precondition rather than leaving it as advice. Carrying ρ in the posterior costs 0.4 experiments against an oracle told its value, recovers 99.3% of recipes where the shipped calibration recovers 2.7%, and beats a five-experiment pilot on both cost and accuracy — so the operational rule simplifies to: do not fix ρ, infer it. The same construction extended to the fall-off shape recovers the shape with near-certainty, and the arms that fix the shape fail under it exactly as the arms that fixed ρ failed before. The pattern is the same one three times over, and it is the paper's most portable finding: *every parameter this family of designers fixes is a silent failure waiting for a system that disagrees, and the remedy each time is to make it latent rather than to calibrate it better.*

What these experiments specify. Read together, the results are less a study of one historical screen than a specification for the evaluator that any such search runs inside. A search procedure consumes an evaluator, and the questions this paper answers are the ones an evaluator must satisfy before searching inside it pays: its gap response must be accurate to within about a factor of two, with under-estimation the safer error (Section 2.4); it must expose uncertainty over structure, not only a score, because a model that cannot represent the right recipe size fails silently at 0.999 confidence (Section 2.5); and its stopping confidence is trustworthy only to the extent its own model is (Section 2.6). Those are testable requirements, and they apply to a learned model of cell state as much as to the calibrated oracle used here.

The measured data adds a fourth requirement and relaxes one assumption. An evaluator fitted to expression data will present a graded, near-additive landscape rather than the sharp threshold the 2006 assay suggests (Section 2.7), which is easier to search, not harder — but it also means the gap response has to be estimated from a pilot rather than inherited from a published figure.

Where the method ends and another begins. The scaling result marks a boundary that matters beyond this benchmark, though not the boundary we first drew. Exact enumeration over hypotheses is what makes the posterior cheap and it is also what fails: at 60 candidates the design step costs 89 s per run even with the structured proposals, and the hypothesis count grows as n^4. A reprogramming search over the roughly 1,600 human transcription factors is three orders of magnitude past that, so the approach used here cannot be carried to it by tuning. Section 2.3 is suggestive about what replaces it: the single change that recovered logarithmic-order growth was to generate proposals from the current posterior's structure rather than draw them at random, which is what population-based search does by construction — a set of candidates derived from the currently promising ones. The natural successor is therefore a search whose proposals are generated, whose posterior is approximate rather than enumerated, and whose evaluator is a learned model, with the requirements above as its acceptance criteria.

Next steps. Four follow from the results, in order of what each would settle. First, replace the calibrated oracle with a retrospective one built directly from measured combinatorial data, turning every count reported here from a property of a model into a measurement. Second, rebuild the proposal distribution along the lines the classical comparator suggests and test whether logarithmic scaling is recovered, which is the difference between diagnosing the defect and repairing it. Third, make the gap response a latent parameter estimated during the search rather than a fixed constant, which the real-data audit shows is the assumption most likely to be wrong in deployment. Fourth, and most open: the threshold-group-testing literature parameterises its designs by a *known* threshold, while the quantity an experimenter lacks is exactly that threshold. Bounds for threshold group testing with unknown u would place the cardinality result of Section 2.5 on a theoretical footing rather than an empirical one.

Limitations. The oracle is a simulator calibrated to published summary figures, so its numbers are properties of a stated model; the design of the comparisons, not the realism of the oracle, carries the conclusions. The audit datasets read out expression rather than a rare phenotype, which is the readout the historical assay used; the evaluator-quality lesson of Okita et al. [26], where a better selection marker changed which colonies counted as success, is the biological form of the same concern. The historical protocol is reconstructed from the paper's description rather than laboratory records. The latent-cardinality designer widens its assumption to sizes 3–6 rather than eliminating it, so a seven-factor truth would reproduce the original failure. Exact enumeration limits the approach to candidate lists of order 10^2, which excludes the genome-scale setting where a reprogramming screen would benefit most.

4 Methods

4.1 Oracle

A hidden essential set $R \subseteq \{1, \ldots, n\}$, $|R| = k$, is drawn uniformly. For a transduced pool S, the colony count is a Poisson draw,

$$y(S) \sim \text{Poisson}(\lambda(S)), \qquad \lambda(S) = B\,\phi(|S \cap R|)\exp(-\delta \max(0, |S| - k)), \tag{2}$$

with $\phi(k) = 1$, $\phi(k-1) = \rho$ and $\phi(j) = 0$ for $j < k-1$. Calibration targets from Takahashi and Yamanaka [1]: a clean four-factor pool yields about 45 colonies ($B = 45$); a three-of-four pool yields a small number lacking full ES-like morphology ($\rho = 0.10$, about 4 colonies); per-cell efficiency falls as more factors are transduced, which is why the reported 10-factor subset outperformed the full 24 ($\delta = 0.0357$, giving 36 and 22 colonies). In threshold-group-testing terms, $u = k$, $l = k-2$, $g = 1$, and ρ is the gap response.

4.2 Inference

The hypothesis space is every k-subset of the candidates ($\binom{24}{4} = 10{,}626$ in the calibrated case), small enough for an exact posterior with no sampling. The next pool maximises expected information gain [3], $\text{EIG}(S) = H[p(y \mid S)] - \mathbb{E}_h\, H[p(y \mid S, h)]$, over 120–260 random pools of size 2–12 plus pools assembled from the most probable factors. The count distribution depends on a hypothesis only through $|S \cap h|$, which takes $k+1$ values, so likelihood and information gain collapse onto $k+1$ overlap classes and the cost of scoring a proposal is independent of posterior concentration. The search stops when the leading hypothesis exceeds τ (0.95 unless stated) or the budget is exhausted.

For unknown cardinality the space becomes every subset of size 3–6 (189,750 at $n = 24$) and the rate depends on a hypothesis through the pair $(|h|, |S \cap h|)$, giving 22 reachable classes, so the same collapse applies one dimension up.

4.3 Strategies and baselines

Historical protocol. n single-factor experiments, the full pool, n leave-one-out experiments, the surviving 10-factor pool, 10 leave-one-out experiments, and a confirmation: $2n + 13$, non-adaptive. **Adaptive design**, with a uniform prior, and **with a literature prior** up-weighting eight factors (the true ones plus decoys), whose quality is varied over coverage, breadth and strength in a separate experiment. **Random pools with Bayesian updating** is the ablation: identical inference, pools drawn at random. **Half-removal search** is the conjunctive analogue of binary splitting, carrying no model beyond a positivity threshold and no representation of recipe size: half the undecided factors are removed and the pool retested; if it remains positive every removed factor was inessential and the candidate set halves, and if it turns negative the removed half contains an essential factor, found by binary search. Its one hyperparameter, the positivity threshold, is the geometric mean of the assumed full and gap rates at the pool size tested, so it carries the same calibration error as the Bayesian designer. **Two-stage cardinality estimation** spends a fixed budget (6 or 12 experiments) estimating the count, then commits to it [21]. **Latent-cardinality design** carries size and identity in one posterior.

Proposal generators. The published generator draws pools uniformly at sizes 2–12 and appends short prefixes of the marginal factor ranking. The structured generator of Section 2.3 returns the same number of pools and adds: removal pools, formed by deleting a random half, quarter or eighth of the factors with posterior mass above 10^{-4}; the complement of each ranked prefix as well as the prefix; and random pools whose sizes are log-uniform on $[d, n]$. Expected information gain

is computed as before, over the full posterior. Scoring a pool requires the overlap $|S \cap h|$ for every hypothesis; because $|S \cap h| = |h| - |\bar{S} \cap h|$, this is counted over whichever side of the pool is smaller, which bounds the work at $n/2$ columns and is exact.

4.4 Latent gap response

The joint state is a $|\mathcal{H}| \times |\mathcal{M}|$ array over recipes and assay models, each model a (shape, ρ) pair. ρ is gridded at $\{0.02, 0.05, 0.10, 0.20, 0.35, 0.55, 0.80, 1.00, 1.25\}$, spaced geometrically below 1 because the likelihood is far more sensitive to a factor-of-two error at 0.05 than at 0.8, and extending past 1 because 29% of measured Norman pairs are sub-additive. The prior is uniform over the grid: seeding it with the measured distribution would assume most of the answer. Likelihood and expected information gain are computed over $(|S \cap h|, m)$ classes as in Section 4.2. Credible intervals for ρ run between grid-cell boundaries rather than between grid points, since the estimand is continuous while the posterior is discrete, and an interval reported as $[0.80, 0.80]$ would exclude a true 0.77 by construction. **Pilot baseline**: the complete reference recipe and each of its one-short pools, $\hat{\rho}$ being the ratio of mean one-short to full yield, at a cost of $d+1 = 5$ experiments counted inside the arm's total. In this benchmark only the hidden recipe yields colonies, so the pilot must be pointed at it; a real calibration would use a previously solved system, so the arm pays a pilot's cost without its risk of calibrating on the wrong one and is an upper bound on what calibration can deliver.

4.5 Measurement-grounded oracles

Empirical gap oracle. Identical to the colony oracle except that ρ is drawn per replicate from the 131 one-short ratios measured in Norman et al. [22] and held fixed across arms within a replicate. Two fall-off shapes are run: *gap*, in which a pool one factor short yields ρ and anything less yields nothing, and *geometric*, in which efficiency is $\rho^{d-|S\cap R|}$. Carrying ρ from a transcriptional response magnitude to a colony rate assumes response magnitude is monotone in assay yield. That assumption is not testable with this data and is the hinge on which Section 2.8 turns.

Retrospective oracle. Answers only for conditions Norman et al. [22] profiled — 105 singles, 131 pairs, one control — and raises an error otherwise, so no value is ever interpolated. The phenotype is $\|x_p - x_0\|$, chosen for being untuned: a score selected to reward a particular biology would make the search succeed by construction. Observation noise is multiplicative lognormal whose scale is the dataset's own split-half discrepancy, estimated by scoring every condition on two random halves of the gene panel, rescaling, and taking the median relative difference (0.049). The surrogate arm fits log pair response on its two constituents' log single responses by ridge regression with the constituents entered in sorted order, and selects the unprofiled condition maximising $\mu + 1.5\sigma$, with σ inflated for conditions whose singles are not yet profiled.

4.6 Campaign and statistics

Results come from independent replicates recorded individually, so that all aggregation is reproducible without re-simulation; strategies within an experiment share hidden recipes and noise seeds, making every comparison paired. Cost is summarised by medians with interquartile ranges, because experiment counts are integer, bounded below and right-skewed. Median intervals are percentile bootstrap (20,000 resamples); proportions use Wilson score intervals, which retain width at 0% and 100%. Paired cost comparisons use a two-sided paired permutation test on the mean difference (20,000 sign flips); paired accuracy comparisons use an exact McNemar test. Success thresholds are estimated by logistic regression in $\log_{10}$ of the varied parameter and reported only when the crossing lies inside the tested range. The statistics layer is implemented in NumPy and checked

against closed-form or planted-truth answers, including the permutation test's false-positive rate under the null.

4.7 Measured datasets

Norman et al. [22] was obtained from the scPerturb harmonisation [25] (`NormanWeissman2019_filtered.h5ad`) and Joung et al. [23] from its combinatorial subset. Counts were normalised to 10^4 per cell, log1p transformed, restricted to the 2000 most variable genes, and averaged per perturbation over conditions with at least 20 cells, giving 237 and 52 conditions respectively. In the hESC subset GFP is an inert partner, so `X+GFP` is the single for X; no GFP-only condition was profiled, so three references are reported, including one fitted by least squares that is deliberately biased toward additivity.

Data and code availability

All code, the per-run records, the analysis layer and its tests, and scripts regenerating every figure are available at `https://github.com/mashal-alkhalifah/yamanaka-replay` under the MIT licence. The headline comparison reproduces in under a minute on two CPU cores with NumPy as the only dependency; the full campaign takes approximately two hours. The two measured datasets are public and the preparation script is included.

Acknowledgements

The code and analyses were implemented with the assistance of Claude (Anthropic) under the direction of the author, who designed the study, specified the experiments and is responsible for all claims and interpretations.

References

[1] Kazutoshi Takahashi and Shinya Yamanaka. Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors. *Cell*, 126(4):663–676, 2006. doi: 10.1016/j.cell.2006.07.024.

[2] Kazutoshi Takahashi, Koji Tanabe, Mari Ohnuki, Megumi Narita, Tomoko Ichisaka, Kiichiro Tomoda, and Shinya Yamanaka. Induction of pluripotent stem cells from adult human fibroblasts by defined factors. *Cell*, 131(5):861–872, 2007. doi: 10.1016/j.cell.2007.11.019.

[3] Dennis V. Lindley. On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics*, 27(4):986–1005, 1956.

[4] Kathryn Chaloner and Isabella Verdinelli. Bayesian experimental design: a review. *Statistical Science*, 10(3):273–304, 1995.

[5] Tom Rainforth, Adam Foster, Desi R. Ivanova, and Freddie Bickford Smith. Modern bayesian experimental design. *Statistical Science*, 39(1), 2024. doi: 10.1214/23-STS915.

[6] Jiaqi Zhang, Louis Cammarata, Chandler Squires, Themistoklis P. Sapsis, and Caroline Uhler. Active learning for optimal intervention design in causal models. *Nature Machine Intelligence*, 5(10):1066–1075, 2023. doi: 10.1038/s42256-023-00719-0.

[7] Christopher Tosh, Mauricio Tec, et al. A Bayesian active learning platform for scalable combination drug screens. *Nature Communications*, 16:156, 2025. doi: 10.1038/s41467-024-55287-7.

[8] Robert Dorfman. The detection of defective members of large populations. *The Annals of Mathematical Statistics*, 14(4):436–440, 1943.

[9] Matthew Aldridge, Oliver Johnson, and Jonathan Scarlett. Group testing: An information theory perspective. *Foundations and Trends in Communications and Information Theory*, 15 (3-4):196–392, 2019. doi: 10.1561/0100000099.

[10] Peter Damaschke. Threshold group testing. In *General Theory of Information Transfer and Combinatorics*, volume 4123 of *Lecture Notes in Computer Science*, pages 707–718. Springer, 2006.

[11] Chun Lam Chan, Sheng Cai, Mayank Bakshi, Sidharth Jaggi, and Venkatesh Saligrama. Near-optimal stochastic threshold group testing, 2013.

[12] Thach V. Bui, Yeow Meng Chee, and Van Khu Vu. Efficient designs for threshold group testing without gap, 2024.

[13] Lorenzo Talamanca and Julian Trouillon. Combinatorial group testing for efficient scaling across biological applications. *Nature Communications*, 17:9146, 2026. doi: 10.1038/s41467-026-77055-5.

[14] Carl Edwards, Edward De Brouwer, Xiner Li, Namkyeong Lee, Ehsan Hajiramezanali, Anne Biton, Sara Mostafavi, and Gabriele Scalia. Biology-in-the-loop: Amortized adaptive hit discovery in CRISPR screens, 2026.

[15] Jason Qin, Hans-Hermann Wessels, Carlos Fernandez-Granda, and Yuhan Hao. Active learning for efficient discovery of optimal gene combinations in the combinatorial perturbation space. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2411.12010.

[16] Owen J. L. Rackham et al. A predictive computational framework for direct reprogramming between human cell types. *Nature Genetics*, 48(3):331–335, 2016. doi: 10.1038/ng.3487.

[17] TFcomb identifies transcription factor combinations for cellular reprogramming based on single-cell multiomics data. *Genome Research*, 35(6):1429–1439, 2025. doi: 10.1101/gr.279955.124. Author list not reproduced here: the article's own pages print contributions by initials only.

[18] Yusuf Roohani, Kexin Huang, and Jure Leskovec. GEARS: Predicting transcriptional outcomes of novel multi-gene perturbations, 2022. bioRxiv preprint; later published in Nature Biotechnology.

[19] Charlotte Bunne, Yusuf Roohani, Yanay Rosen, Ankit Gupta, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities. *Cell*, 187(25):7045–7063, 2024. doi: 10.1016/j.cell.2024.11.015.

[20] Jia Bi, Samuel Pinilla, and Chenyang Zhu. Held-out evidence resolves follow-up measurement decisions in biological screens, 2026.

[21] Nader H. Bshouty, Vivian E. Bshouty-Hurani, George Haddad, Thomas Hashem, Fadi Khoury, and Omar Sharafy. Adaptive group testing algorithms to estimate the number of defectives. In *Proceedings of Algorithmic Learning Theory (ALT)*, volume 83 of *Proceedings of Machine Learning Research*, pages 93–110, 2018.

[22] Thomas M. Norman, Max A. Horlbeck, Joseph M. Replogle, Alex Y. Ge, Albert Xu, Marco Jost, Luke A. Gilbert, and Jonathan S. Weissman. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science*, 365(6455):786–793, 2019. doi: 10.1126/science.aax4438.

[23] Julia Joung, Sai Ma, Tristan Tay, et al. A transcription factor atlas of directed differentiation. *Cell*, 186(1):209–229.e26, 2023. PMID 36608654.

[24] Joseph M. Replogle, Reuben A. Saunders, Angela N. Pogson, Jeffrey A. Hussmann, et al. Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq. *Cell*, 185(14):2559–2575.e28, 2022.

[25] Stefan Peidli, Tessa D. Green, Ciyue Shen, Torsten Gross, et al. scPerturb: harmonized single-cell perturbation data. *Nature Methods*, 2024. doi: 10.1038/s41592-023-02144-y.

[26] Keisuke Okita, Tomoko Ichisaka, and Shinya Yamanaka. Generation of germline-competent induced pluripotent stem cells. *Nature*, 448:313–317, 2007. doi: 10.1038/nature05934.